



8657 - How does Rotation Impact Y Dwarf Atmospheres?

Cycle: 4, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

1D radiative-convective equilibrium models systematically disagree with observations of brown dwarfs and exoplanets cooler than 700 K. These model shortcomings hinder efforts to interpret JWST data and suggest a gap in our current understanding of cold H₂-He dominated atmospheres, one of JWST's premiere discovery spaces. We propose to investigate a promising cause of the disagreement-- rotational modulation of temperature structures, circulation patterns, chemistry, clouds, and emergent spectra. Our project will determine whether accounting for rotation can resolve data-model discrepancies and whether rotation induces observable latitude-dependent changes to atmospheric properties in three phases: (1) Run convection-resolving circulation models with chemical tracers near the inferred properties of the two coldest objects with known rotation periods (W0359, W1405), and test corresponding synthetic spectra against their archival data. (2) Run a small grid of these simulations varying temperature, surface gravity, rotation rate, and clear vs cloudy in order to map out how effects scale more generally. (3) Construct a large grid of adjusted RCE models with latitude and rotation rate included along with the usual axes of temperature, surface gravity and composition. Build tools to combine them into spectra for pseudo-2D models at different viewing angles. Test these on the GO2302 dataset (panchromatic spectra for 18 objects cooler than 700 K), then make them publicly available, enabling JWST-users to quickly adopt our findings into their own analyses.

OBSERVING DESCRIPTION

n/a archival theory